

## SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO2:** Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)

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**QUESTIONS: 5**

**Total Marks:50**

Annexure B  
Analytical Problem Solving

| <b>Criteria</b>  | <b>Problem Understanding (2)</b> | <b>Method Selection (2)</b> | <b>Solution Process (2.5)</b> | <b>Accuracy of Calculation (2)</b> | <b>Interpretation of Results (1.5)</b> | <b>Total(10)</b> |
|------------------|----------------------------------|-----------------------------|-------------------------------|------------------------------------|----------------------------------------|------------------|
| <i>Weightage</i> | 20%                              | 20%                         | 25%                           | 20%                                | 15%                                    | 100%             |
| <i>Q1</i>        |                                  |                             |                               |                                    |                                        |                  |
| <i>Q2</i>        |                                  |                             |                               |                                    |                                        |                  |
| <i>Q3</i>        |                                  |                             |                               |                                    |                                        |                  |
| <i>Q4</i>        |                                  |                             |                               |                                    |                                        |                  |
| <i>Q5</i>        |                                  |                             |                               |                                    |                                        |                  |

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***Code of Conduct:***

*I Ch.Sathish / 192311032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: Ch.Sathish***

## RUBRICS FOR CALCULATION:

| Numerical Problems        |           |                                                                                   |                                           |                                            |                                               |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Criteria                  | Weightage | Level 4<br>(Exemplary)                                                            | Level 3<br>(Proficient)                   | Level 2<br>(Developing)                    | Level 1<br>(Beginning)                        |
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear processes                            |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |

## Annexure B Analytical Problem Solving

**1. Data Cleaning using Duplicate Detection (Patient Registration) (5 Marks)** The following patient IDs were extracted from a registration database.

P101, P102, P105, P103, P105, P107, P108, P103, P110, P112

Perform the following:

1. Identify duplicate records.
2. Remove duplicate entries.
3. Determine the number of unique patients.

**2. Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)**

The following data represent treatment cost (₹ thousand) and hospital stay (days).

| Treatment Cost (x) | Hospital Stay (y) |
|--------------------|-------------------|
| 20                 | 3                 |
| 25                 | 4                 |
| 30                 | 5                 |
| 35                 | 6                 |
| 40                 | 8                 |

Perform the following:

1. Compute the mean of x and y.
2. Calculate the covariance.
3. Interpret the relationship between treatment cost and duration of stay.

**3. Outlier Detection using Z-Score (Patient Recovery Time) (10 Marks)** Recovery time (days) for patients after surgery is recorded as:

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35

Perform the following:

1. Compute the mean.
2. Compute the standard deviation.
3. Calculate the z-score for each observation.
4. Identify the outlier(s) using the criterion  $|z| > 3$ .

**4. Data Discretization using Equal-Width Binning (Blood Glucose Levels) (5 Marks)** A clinic records fasting blood glucose levels (mg/dL).

Sorted Data:

75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145

Partition the data into 3 equal-width bins.

**Replace each observation by the nearest bin boundary. Write the transformed dataset.**

**5. Standardization of Insurance Claim Amounts**  
**(10 Marks)** An insurance company records claim amounts (₹ thousand): 12, 15, 18, 20, 22, 25, 28, 30, 35, 40

**Perform the following:**

- 1. Compute the mean.**
- 2. Compute the standard deviation.**
- 3. Calculate the z-score for every observation.**
- 4. Identify claims that are above one standard deviation from the mean.**

## Answers

### 1. Data Cleaning using Duplicate Detection (Patient Registration)

**Sol:** Given Patient IDs:

P101, P102, P105, P103, P105, P107, P108, P103, P110, P112

#### Step 1: Identify Duplicate Records

The duplicate patient IDs are:

- P105
- P103

#### Step 2: Remove Duplicate Entries

After removing duplicates, the unique patient IDs are:

P101, P102, P105, P103, P107, P108, P110, P112

#### Step 3: Determine the Number of Unique Patients

Total unique patient IDs: 8

### Final Answer

- Duplicate Records: P105, P103
- Dataset after Removing Duplicates: P101, P102, P105, P103, P107, P108, P110, P112
- Number of Unique Patients: 8

### 2. Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)

The following data represent treatment cost (₹ thousand) and hospital stay (days).

**Sol:**



1.

**Step 1: Compute the Mean of x and y**

Mean of Treatment Cost (x):

$$\bar{x} = \frac{20 + 25 + 30 + 35 + 40}{5} = \frac{150}{5} = 30$$

Mean of x = 30 (₹ thousand)

Mean of Hospital Stay (y):

$$\bar{y} = \frac{3 + 4 + 5 + 6 + 8}{5} = \frac{26}{5} = 5.2$$

Mean of y = 5.2 days

**Step 2: Calculate the Covariance**

| x  | y | $x - \bar{x}$ | $y - \bar{y}$ | $(x - \bar{x})(y - \bar{y})$ |
|----|---|---------------|---------------|------------------------------|
| 20 | 3 | -10           | -2.2          | 22                           |
| 25 | 4 | -5            | -1.2          | 6                            |
| 30 | 5 | 0             | -0.2          | 0                            |
| 35 | 6 | 5             | 0.8           | 4                            |
| 40 | 8 | 10            | 2.8           | 28                           |

Sum of products =  $22 + 6 + 0 + 4 + 28 = 60$

Covariance (using  $n = 5$ ):

$$\text{Cov}(x, y) = \frac{60}{5} = 12$$

Covariance = 12

**Step 3: Interpretation**

Since the covariance is positive (12), there is a positive relationship between treatment cost and hospital stay.

Interpretation: As the treatment cost increases, the hospital stay tends to increase.

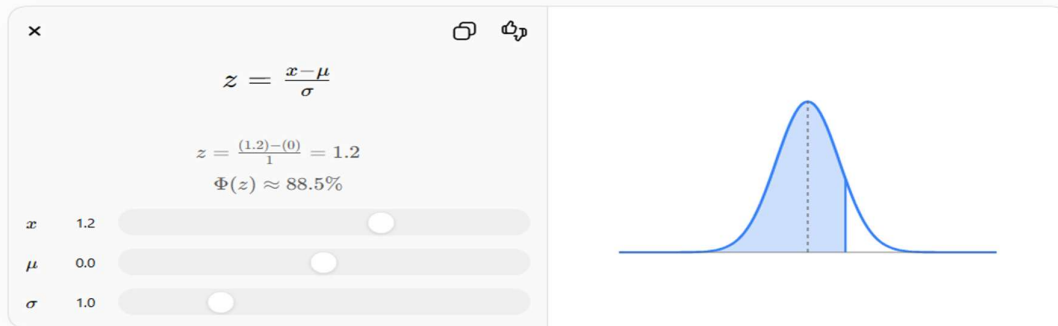
**3. Outlier Detection using Z-Score (Patient Recovery Time) (10 Marks)**

Recovery time (days) for patients after surgery is recorded as:

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35

Given Data (Recovery Time in Days):

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35



**Sol:**

**1.**

**Sum of observations:**

$$8 + 9 + 10 + 10 + 11 + 12 + 12 + 13 + 14 + 15 + 35 = 149$$

Number of observations (n) = 11

$$\text{Mean} = \frac{149}{11} = 13.55$$

Mean = 13.55 days

**Step 2: Compute the Standard Deviation**

Using the population standard deviation formula:

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$
$$\sigma \approx 7.25$$

Standard Deviation = 7.25 days

**Step 3: Calculate the Z-Score**

$$z = \frac{x - \bar{x}}{\sigma}$$

| Recovery Time<br>(x) | Z-Score |
|----------------------|---------|
| 8                    | -0.77   |
| 9                    | -0.63   |
| 10                   | -0.49   |
| 10                   | -0.49   |

| Recovery Time (x) | Z-Score |
|-------------------|---------|
| 11                | -0.35   |
| 12                | -0.21   |
| 12                | -0.21   |
| 13                | -0.08   |
| 14                | 0.06    |
| 15                | 0.20    |
|                   |         |

#### Step 4: Identify the Outlier(s)

Criterion:

$$|z| > 3$$

The largest z-score is:

- $35 \rightarrow z = 2.96$

Since  $2.96 < 3$ , there are no outliers according to the given criterion.

#### 4. Data Discretization using Equal-Width Binning (Blood Glucose Levels) (5 Marks)

Sol: Given Sorted Data (mg/dL):

75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145

##### Step 1: Calculate Bin Width

- Minimum value = 75
- Maximum value = 145
- Number of bins = 3

$$\text{Bin Width} = \frac{145 - 75}{3} = \frac{70}{3} \approx 23.33$$

##### Step 2: Form Equal-Width Bins

- Bin 1: 75 – 98.33
- Bin 2: 98.34 – 121.66
- Bin 3: 121.67 – 145

##### Step 3: Replace Each Observation by the Nearest Bin Boundary

| Original Value | Nearest Boundary | Replaced Value |
|----------------|------------------|----------------|
| 75             | 75               | 75             |
| 82             | 75               | 75             |
| 88             | 98.33            | 98.33          |
| 92             | 98.33            | 98.33          |

| Original Value | Nearest Boundary | Replaced Value |
|----------------|------------------|----------------|
| 95             | 98.33            | 98.33          |
| 99             | 98.34            | 98.34          |
| 104            | 98.34            | 98.34          |
| 110            | 121.66           | 121.66         |
| 118            | 121.66           | 121.66         |
| 122            | 121.67           | 121.67         |
| 130            | 121.67           | 121.67         |
| 145            | 145              | 145            |

#### Step 4: Transformed Dataset

75, 75, 98.33, 98.33, 98.33, 98.34, 98.34, 121.66, 121.66, 121.67, 121.67, 145

#### 5. Standardization of Insurance Claim Amounts (10 Marks)

An insurance company records claim amounts (₹ thousand):

12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Perform the following:

Sol: Given Claim Amounts (₹ thousand):



#### Step 1: Compute the Mean

Sum of observations:

$$12 + 15 + 18 + 20 + 22 + 25 + 28 + 30 + 35 + 40 = 245$$

Number of observations (n) = 10

$$\text{Mean} = \frac{245}{10} = 24.5$$

**Mean = 24.5 (₹ thousand)**

Step 2: Compute the Standard Deviation

Using the population standard deviation formula:

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$
$$\sigma \approx 8.20$$

Standard Deviation = 8.20 (₹ thousand)

**Step 3: Calculate the Z-Score**

$$z = \frac{x - \bar{x}}{\sigma}$$

| Claim Amount (₹ thousand) | Z-Score |
|---------------------------|---------|
| 12                        | -1.52   |
| 15                        | -1.16   |
| 18                        | -0.79   |
| 20                        | -0.55   |
| 22                        | -0.30   |
| 25                        | 0.06    |
| 28                        | 0.43    |
| 30                        | 0.67    |
| 35                        | 1.28    |
| 40                        | 1.89    |

**Step 4: Identify Claims Above One Standard Deviation**

**Criterion:**

$$z > 1$$

The claim amounts with z-score  $> 1$  are:

- ₹35 thousand ( $z = 1.28$ )
- ₹40 thousand ( $z = 1.89$ )